

```

module peak_detector (
    input clk,
    input rst,
    input [7:0] data_in,
    output reg peak
);

    reg [7:0] prev, curr, next;

    always @(posedge clk or posedge rst) begin
        if (rst) begin
            prev <= 0;
            curr <= 0;
            next <= 0;
            peak <= 0;
        end else begin
            prev <= curr;
            curr <= next;
            next <= data_in;

            if ((curr > prev) && (curr > next))
                peak <= 1;
            else
                peak <= 0;
        end
    end

endmodule

```